

Lecture 16

Least Squares Problem I

Instructors:

L01 Cosme Louart 程高木

L02 Junfeng Wu 吴均峰

L03&L04 Ruoyu Sun 孙若愚



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

数据科学学院

School of Data Science

Part 0

Roadmap and Outline

3 mins



Roadmap of MATH2041

Segment 1 Solving linear systems (Lec 3-15)

Segment 2 [Lec 16-20]: Three relatively independent parts:

- Solving [least squares problem](#) (Lec 16,17)
- Determinant. [Important tool!] (Lec 18)
- [Linear transformation](#). [Advanced math perspective of [matrix](#)] (Lec 19,20)

Segment 3 Lec 21-27: Eigenvalues and related.

- [Eigenvalues](#). Lec 21-24
- [Singular values](#). Lec 25-26
- Quadratic forms. Lec 27.

Today's Lecture: Outline

Today: Least Squares Problem

1. Motivation: Linear Regression
2. Least Squares Problem
3. Solving Least Squares Problem

After today's lecture, you shall be able to:

1. Model certain real-world problems as linear regression problems
2. Tell the relation/difference of least squares and linear equations
3. Solve a least squares problem

Roadmap of Linear Systems: Extension

Theme: Solving
Linear Systems

Math Tools

What is Learned?
(On linear system)

Part 1 Preparation

Lec 3,4,5

Matrix multiplication
Row operations

Linear system: $Ax = b$

Part 2 Solving
Square system

Lec 6,7,8,9,10

LU Decomposition
Inverse

“Good” square system:
Solution $A^{-1}b$

Part 3 Solving
rectangular system

Lec 11,12, 13

Linear space
Span

Rectangular system:
Solution $x_p + N(A)$

Part 4 Solution set
structure

Lec 13,14,15

Linear independence
Basis
Dimension
Rank

Solution set dim $(n-r)$
Column space dim r

Method &
Analytical
Expressions

Deeper
understanding

Part 5 Approximate
solutions

Lec 16,17

Least square

Find an approximate solution

Today's topic

Part I Motivation: Linear Regression

15 mins

Motivation: Salary Prediction

Example (Salary prediction)

	Score	Assist	Salary (million dollars)
Paul	16	10	35
Harden	25	8	45
Capela	18	3	15
Tom	12	14	???

Suppose I'm the Boss of Rocket.
Now I want to hire a new guy Tom.

How much shall I pay Tom?

Motivation: Salary Prediction

	Score	Assist	Salary (million dollars)
Paul	16	10	35
Harden	25	8	45
Capela	18	3	15
Tom	12	14	???

Linear model:

Assume Salary = $x_1 \cdot \text{Score} + x_2 \cdot \text{Assist}$.

Equations:

$$\begin{cases} 35 = 16x_1 + 10x_2 \\ 45 = 25x_1 + 8x_2 \\ 15 = 18x_1 + 3x_2 \end{cases}$$

$$\begin{bmatrix} 16 & 10 \\ 25 & 8 \\ 18 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 35 \\ 45 \\ 15 \end{bmatrix}$$

Is there a solution? Try to solve it.

How to Proceed?

Is there a solution?

But....Business is not math class. You cannot say “no solution”.
You have to find "a solution". So... what to do?

How to Proceed?

Is there a solution?

But.... Business is not math class. You cannot say “no solution”.
You have to find "a solution". So... what to do?

Find an **approximate** solution!

Find x s.t.

$$\begin{cases} 16x_1 + 12x_2 \approx 35 \\ 25x_1 + 8x_2 \approx 45 \\ 18x_1 + 3x_2 \approx 15 \end{cases}$$

$$Ax \approx b$$

“Approximate solution” means
“**error**” is small.

What error?

Error 1: $\epsilon_1 = 16x_1 + 12x_2 - 35$

Error 2: $\epsilon_2 = 25x_1 + 8x_2 - 45$

Error 3: $\epsilon_3 = 18x_1 + 3x_2 - 15$

$$|\epsilon_1| + |\epsilon_2| + |\epsilon_3|.$$

Single metric?

$$\sqrt{\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2} = \left\| \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix} \right\| = \|Ax - b\|$$

(“row-form”).

What Approximate Solution?

Find an approximate solution!

Find x s.t. $\|Ax - b\|$ is small_____

Still not clear enough: how small is small?

Analogous setting:

Shenzhen Basketball coach asks a teacher:

I want to find a tall person in your class.

What does “tall person” mean? 1.9m? 1.95m?

Instead, find the “tallest person” in your class.
Let the coach decide.

What Approximate Solution?

Find an approximate solution!

Find x s.t. $\|Ax - b\|$ is small_____

One trial: Set the goal as find x s.t. $\|Ax - b\|^2 \leq 10^{-8}$

Issue: Such x may not exist.

Anyways, Boss needs "a solution".

Method: Make error as small as possible.

Let the Boss decide whether it's acceptable.

Minimization Problem

Method: Make error as small as possible.

Find $\underline{x}$ s.t. $\|Ax - b\|$ is the smallest among all.

More precisely, find $\underline{x}^*$ such that

$$\|Ax^* - b\| \leq \|Ax - b\| \quad \forall x \in \mathbb{R}^n$$

Equivalently, solve the following problem:

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|$$

Part II Least Squares

Definition

Row Interpretation: Linear Regression

17 mins

Least Squares

What if a linear system $A\mathbf{x} = \mathbf{b}$ has no solution?

One method: Find the best “approximation”!

Definition (Least Squares Problem)

Given $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^{m \times 1}$, the **least squares problem** is

$$\min_{\mathbf{x} \in \mathbb{R}^{n \times 1}} \|A\mathbf{x} - \mathbf{b}\| \quad (*)$$

Remark: $\|\cdot\|$ is the **Euclidean norm** of vectors (lecture 2)

$\mathbf{y}$ is a **solution** of the problem (*) (or “the least square problem”) iff

$$\|A\mathbf{y} - \mathbf{b}\| \leq \|A\mathbf{x} - \mathbf{b}\| \text{ for any } \mathbf{x}$$

Do not call this $\mathbf{y}$ a “solution of the linear system $A\mathbf{x} = \mathbf{b}$ ”!

Relation of Least Squares & Linear Systems

minimal value

1) If $\exists \mathbf{x}$ s.t. $\mathbf{Ax} = \mathbf{b}$, then $\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\| = 0$,
and solving $\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\| \iff$ solving $\mathbf{Ax} = \mathbf{b}$.

2) If $\nexists \mathbf{x}$ s.t. $\mathbf{Ax} = \mathbf{b}$, then the two problems are different.

Least Squares

Judgement: True or False?

The two problems are equivalent, in the sense that their

solution sets are the same.

Yes

$$(P1) \min_x \| \underline{Ax - b} \|$$

$$(P2) \min_x \| \underline{Ax - b} \|^2$$

Rewrite what they mean:

$$y \text{ is a } \underline{\text{solution}} \text{ of (P1)} \Leftrightarrow (0 \leq) \|Ay - b\| \leq \|Ax - b\|, \forall x \in \mathbb{R}^n$$

$$\Leftrightarrow \|Ay - b\|^2 \leq \|Ax - b\|^2, \forall x \in \mathbb{R}^n \text{ (manipulate relation)}$$

$$\Leftrightarrow y \text{ is a solution of (P2)}$$

$$(P1) \Leftrightarrow \min_x \|Ax - b\|^3$$

$$\Leftrightarrow \min_x \|Ax - b\|$$

Least Squares: Row Form and Column Form

Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, with:

Column form: $\mathbf{A} = [\vec{a}_1 \quad \vec{a}_2 \quad \dots \quad \vec{a}_n]$

$$\mathbf{A}\mathbf{x} = \chi_1 \vec{a}_1 + \dots + \chi_n \vec{a}_n$$

Row form: $\mathbf{A} =$

$$\begin{bmatrix} \mathbf{a}_{(1)}^\top \\ \mathbf{a}_{(2)}^\top \\ \vdots \\ \mathbf{a}_{(m)}^\top \end{bmatrix}$$

Row form of least squares:

$$\begin{aligned} \underbrace{\|\mathbf{A}\mathbf{x} - \mathbf{b}\|}_{1 \times 1}^2 &= \left\| \begin{pmatrix} \mathbf{a}_{(1)}^\top \mathbf{x} - b_1 \\ \mathbf{a}_{(2)}^\top \mathbf{x} - b_2 \\ \vdots \\ \mathbf{a}_{(m)}^\top \mathbf{x} - b_m \end{pmatrix} \right\|^2 = \left\| \begin{pmatrix} \Sigma_1 \\ \Sigma_2 \\ \vdots \\ \Sigma_m \end{pmatrix} \right\|^2 = \Sigma_1^2 + \dots + \Sigma_m^2 \\ &= \underbrace{(\underbrace{\mathbf{a}_{(1)}^\top}_{1 \times n} \underbrace{\mathbf{x}}_{n \times 1} - \underbrace{b_1}_{1 \times 1})^2}_{\text{residual}} + \dots + (\mathbf{a}_{(m)}^\top \mathbf{x} - b_m)^2 \end{aligned}$$

Interpretation: LS objective is a sum of squares of residual components

so least squares minimizes sum of squares of residuals

Column form of least squares:

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \left\| \underbrace{\chi_1 \vec{a}_1}_{m \times 1} + \dots + \underbrace{\chi_n \vec{a}_n}_{m \times 1} - \underbrace{\vec{b}}_{m \times 1} \right\|^2 = \underbrace{(\underbrace{\chi_1 \vec{a}_1}_{\text{vec}} - \underbrace{b_1}_{\text{scalar}})^2}_{\text{residual}} + \dots + \dots$$

Interpretation: discuss later.

Application of Least Squares: Linear Regression

Examples (Salary prediction)

	Score	Assist	Salary (million dollars)
Paul	16	10	35
Harden	25	8	43
Capela	18	3	16
Tom	12	14	???

How much shall I pay Tom?

Step 1: (estimate the value of “score” and “assist”)

Compute $\mathbf{x}$ s.t. the $\sum_{i=1}^m (b_i - \mathbf{x}^T \mathbf{a}_i)^2 = \|A\mathbf{x} - \mathbf{b}\|^2$ is the smallest [later]

(among all possible choices of $\mathbf{x}$), where $A = \begin{bmatrix} 16 & 10 \\ 25 & 8 \\ 18 & 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 35 \\ 43 \\ 16 \end{bmatrix}$

Denote the solution as $\mathbf{y}$.

Step 2: (estimate salary)

Compute $\mathbf{y}^T \mathbf{a}_{\text{Tom}} = 12y_1 + 14y_2$

Step 2 is the inference stage

Linear Regression is Fundamental

Machine learning:

Linear regression is the first model.

Deep learning (深度学习):

Linear regression is the foundation.

Statistics:

Will spend many lectures on it.

Econometrics (计量经济学):

Will spend many lectures on it.

...

Don't be too surprised (and bored) if you see it again next year.

Part III Solving Least Squares

35 mins

Part III.1 Solving LS is Finding Projection

15 mins

Toy Problem

Find $x \in \mathbb{R}$ s.t. $(x - 1)^2$ is minimized.

Answer: $x = 1$.

How to verify it is a solution? Let's understand the formal process.

The problem is: find $y \in \mathbb{R}$ s.t.

$$(y - 1)^2 \leq (x - 1)^2 \text{ for any } x.$$

Answer: $y = 1$.

Verify:

When $y=1$, $(y-1)^2 = 0 \leq (x-1)^2, \forall x \in \mathbb{R}$.
So $y=1$ is the solution of $\min_{x \in \mathbb{R}} (x-1)^2$.

Reformulation

column form

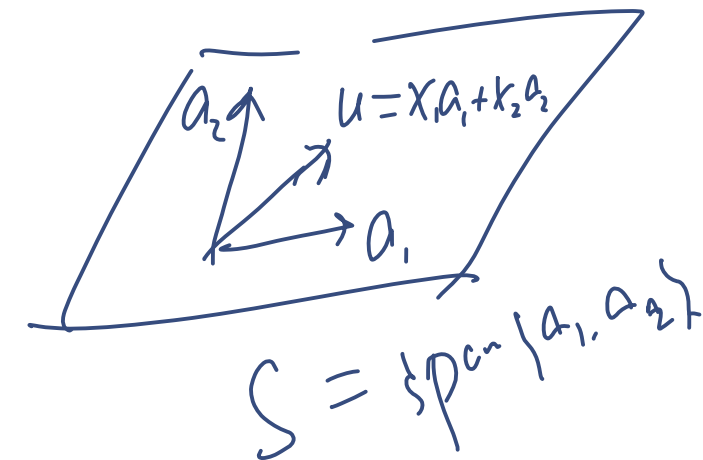
$$\min \|A\mathbf{x} - \mathbf{b}\| = \|x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n - \mathbf{b}\|$$

What if $n = 2$?

$$\text{Find } \mathbf{y} \in \mathbb{R}^2 \text{ s.t. } \|\mathbf{y}_1\mathbf{a}_1 + \mathbf{y}_2\mathbf{a}_2 - \mathbf{b}\| \leq \|\mathbf{x}_1\mathbf{a}_1 + \mathbf{x}_2\mathbf{a}_2 - \mathbf{b}\|, \forall \mathbf{x}. (*)$$

Can replace $x_1\mathbf{a}_1 + x_2\mathbf{a}_2$ by a point in $\text{span}\{\mathbf{a}_1, \mathbf{a}_2\}$, a plane.

$\mathbf{y}_1\mathbf{a}_1 + \mathbf{y}_2\mathbf{a}_2$ also a point in $\text{span}\{\mathbf{a}_1, \mathbf{a}_2\}$.



Problem P1 find $\mathbf{y}$ s.t.

$$\|A\mathbf{y} - \mathbf{b}\| \leq \|A\mathbf{x} - \mathbf{b}\| \text{ for any } \mathbf{x} (*)$$

$$\Leftrightarrow \text{P1'} \text{ Find } \mathbf{p} \in C(A) \text{ s.t. } \|\mathbf{p} - \mathbf{b}\| \leq \|\mathbf{u} - \mathbf{b}\| \text{ for any } \mathbf{u} \in C(A)$$

Example of $n=2$

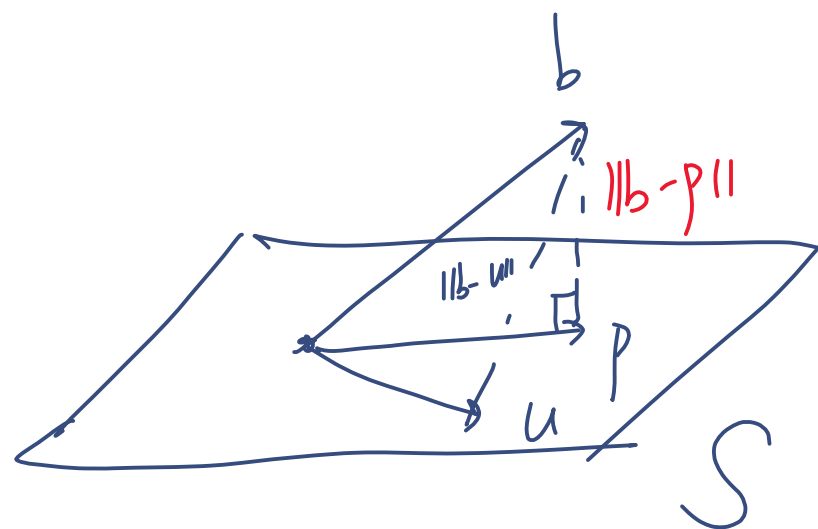
Suggested break

$$\min \|Ax - \mathbf{b}\| = \|x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n - \mathbf{b}\|$$

What if $n = 2$? Find $p \in \text{span}\{\mathbf{a}_1, \mathbf{a}_2\} = S$ s.t. $\|p - \mathbf{b}\| \leq \|u - \mathbf{b}\|, \forall u \in S$.

Find $y \in \mathbb{R}^2$ s.t. $\|y_1 \mathbf{a}_1 + y_2 \mathbf{a}_2 - \mathbf{b}\| \leq \|x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 - \mathbf{b}\|, \forall x. (*)$

Geometry: Draw p , b , u ; Translate $(*)$ to geometry.



$$\|b - p\| \leq \|b - u\|, \forall u \in S.$$

Find a point p on S
whose distance to b is smallest
or closest to b .

Claim: Suppose p satisfies

$$\mathbf{b} - \mathbf{p} \perp S.$$

then $\|b - p\| \leq \|b - u\|, \forall u \in \text{span}(\mathbf{a}_1, \mathbf{a}_2)$.

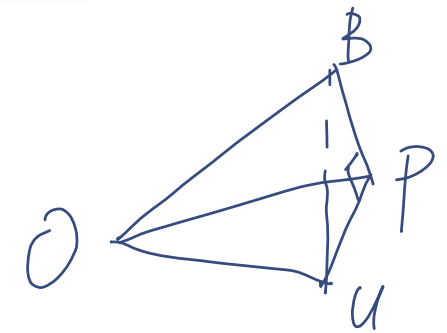
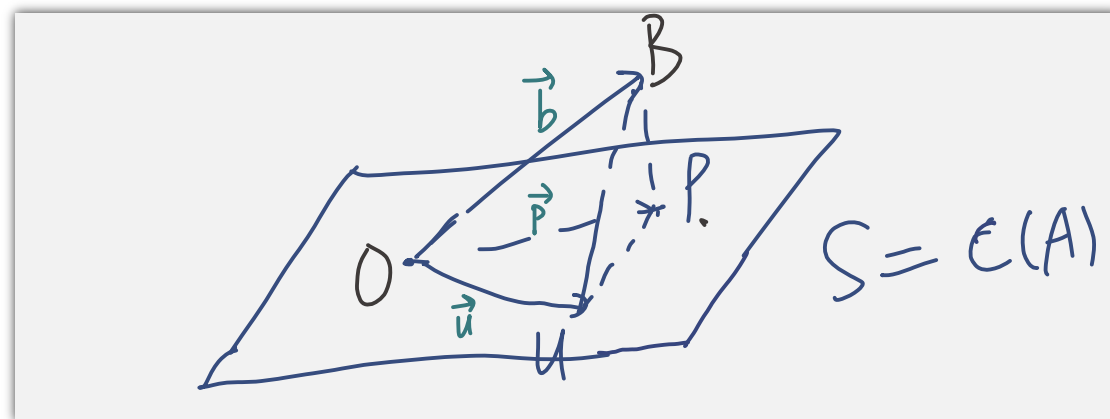
General Case: Geometry

$$\min \|A\mathbf{x} - \mathbf{b}\| = \|x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n - \mathbf{b}\|$$

Let's consider the general case.

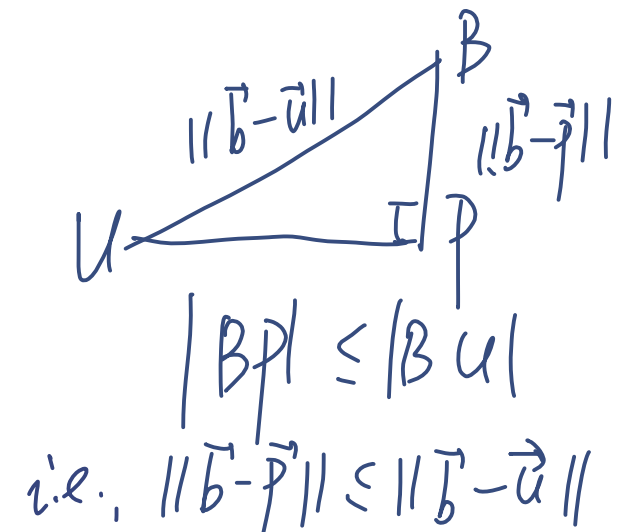
Geometry problem:

Find a point p in $\text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\} = C(A)$, such that the distance to $\mathbf{b}$ is the smallest.



$BP \perp \text{space } S$, so $BP \perp \vec{u}$

Claim: Suppose p satisfies $\vec{b} - \vec{p} \perp S$,
then $\|b - p\| \leq \|b - u\|, \forall u \in \text{span}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$.



Projection: Lemma and Definition

Lemma 16.1 (Orthogonal Projection)

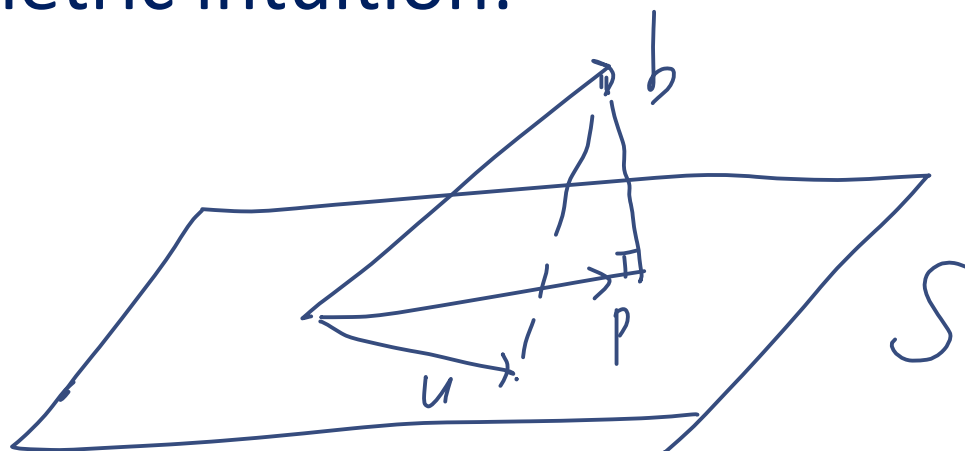
Suppose S is a subspace of $\mathbb{R}^m$. Suppose $\mathbf{p} \in S$, then

$$(1) \|\mathbf{b} - \mathbf{p}\| \leq \|\mathbf{b} - \mathbf{z}\|, \forall \mathbf{z} \in S$$

$\iff$

$$(2) \mathbf{b} - \mathbf{p} \perp S.$$

Proof idea & geometric intuition:



This lemma holds for
any subspace S ,
not just $C(A)$

Formal proof: skipped.

Definition 16.1

Suppose S is a subspace of $\mathbb{R}^m$.

Suppose $\mathbf{p} \in S$ and $\mathbf{b} - \mathbf{p} \perp S$, then we say $\mathbf{p}$ is the **projection of $\mathbf{b}$ onto S** .

Part III.2 Finding Projection is Solving Equation

17 mins

Restate the Problem of Finding p

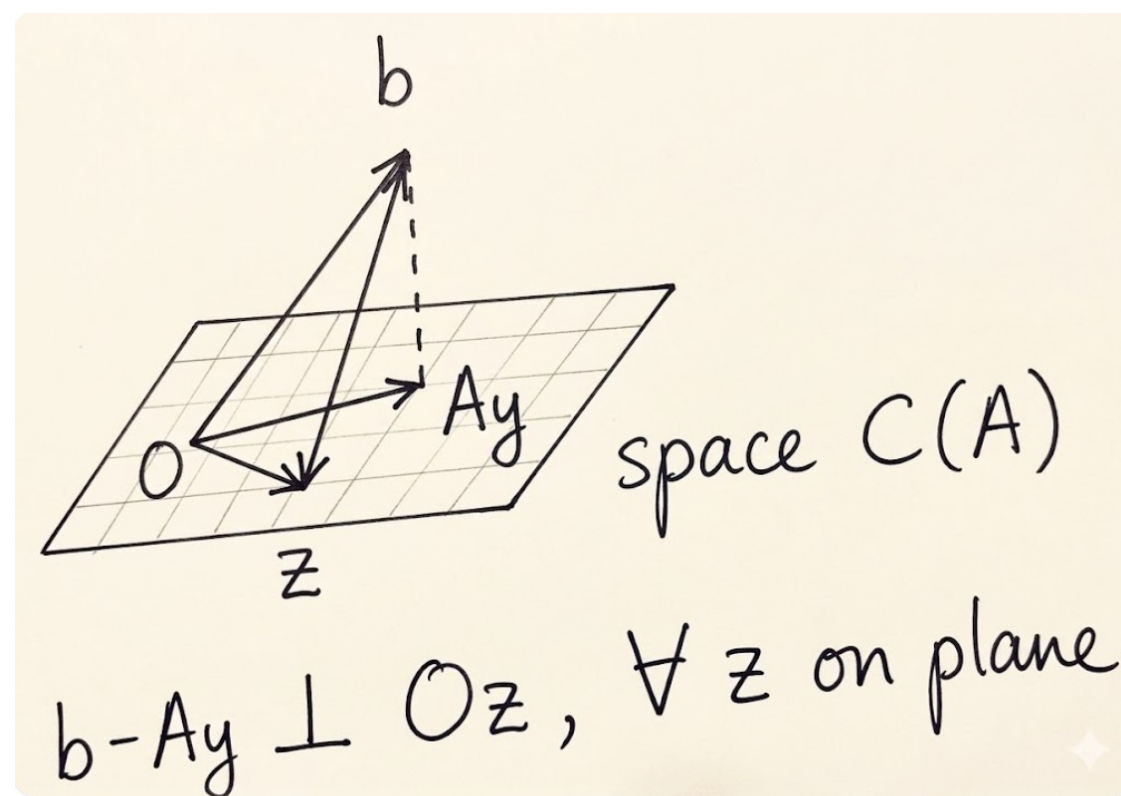
Problem P1 find $\mathbf{y}$ s.t. $\|A\mathbf{y} - \mathbf{b}\| \leq \|A\mathbf{x} - \mathbf{b}\|$ for any $\mathbf{x}$ (*)

$\Leftrightarrow$ **P1'** Find $\mathbf{p} \in S = \mathcal{C}(A)$ s.t. $\|\mathbf{p} - \mathbf{b}\| \leq \|\mathbf{u} - \mathbf{b}\|$ for any $\mathbf{u} \in \mathcal{C}(A)$

Lemma 16. says, just need to solve the following problem:

P2: find $p \in S$ such that $\mathbf{b} - \mathbf{p} \perp S$ (**) [True for any subspace S]

where $S = \mathcal{C}(A)$ [use structure of our problem]



Next: Make It Solvable

Problem P2: Find $p \in C(A)$, s.t. $\mathbf{b} - \mathbf{p} \perp C(A)$ (**)

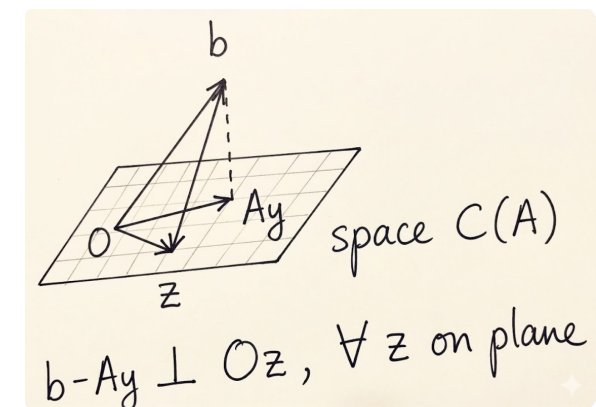
Recall from Lec 13:

Property: $A \in \mathbb{R}^{m \times n}$. Then $C(A) = \{A\vec{x} \mid \vec{x} \in \mathbb{R}^{n \times 1}\}$.

Let's analyze this problem, to make it solvable.

Fact 1: Find some $p \in C(A)$ satisfying xxxx

$\Leftrightarrow$ find some y where Ay satisfies xxxxx
[Property of $C(A)$]



Fact 2: $v \perp C(A) \Leftrightarrow v \perp u, \forall u \in C(A)$ [Def. of "vector $\perp$ space"; Lec 15]

$\Leftrightarrow v \perp Ax, \forall x \in \mathbb{R}^n$ [Property of $C(A)$]

Combining the two facts, the problem P1 is equivalent to:

P2': find $y \in \mathbb{R}^{n \times 1}$ such that

$$\mathbf{b} - A\mathbf{y} \perp A\mathbf{x}, \forall \mathbf{x} \in \mathbb{R}^n.$$

Next: Make It Solvable

Just need to find $\mathbf{y} \in \mathbb{R}^{n \times 1}$ such that

$$\mathbf{b} - \mathbf{A}\mathbf{y} \perp \{\mathbf{A}\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^{n \times 1}\}$$

$$\iff \underbrace{(\mathbf{b} - \mathbf{A}\mathbf{y})^\top}_{1 \times m} \underbrace{\mathbf{A}}_{m \times n} \underbrace{\mathbf{x}}_{n \times 1} = 0, \quad \forall \mathbf{x} \in \mathbb{R}^n, \text{ i.e. } \underbrace{\mathbf{z}^\top}_{1 \times n} \underbrace{\mathbf{x}}_{n \times 1} = 0, \quad \forall \mathbf{x} \in \mathbb{R}^n.$$

$$\iff (\mathbf{b} - \mathbf{A}\mathbf{y})^\top \mathbf{A} = \mathbf{0}$$

[apply Lemma 16.1]

Lemma 16.1:

$$\mathbf{u}^\top \mathbf{x} = 0, \quad \forall \mathbf{x} \in \mathbb{R}^n$$

iff $\mathbf{u} = \mathbf{0}$.

Proved in the next page.

Small Result Used in Last Page

Lemma 16.1: If $z \in \mathbb{R}^n$ satisfies $z \perp x, \forall x \in \mathbb{R}^n$, then $z = 0$.

Proof Method 1 (regular):

Write definition.
Try some x ($n=2$).

$$z^T x = z_1 x_1 + z_2 x_2 = 0, \quad \forall x_1, x_2 \in \mathbb{R}$$

Pick $(x_1, x_2) = (1, 0)$, then $z_1 = 0$,

Pick $(x_1, x_2) = (0, 1)$, then $z_2 = 0$.

Extend to general n .

$$0 = z^T x, \quad \forall x \in \mathbb{R}^n.$$

Pick $x = e_i = [0, \dots, 0, \underset{\substack{\downarrow \\ i\text{-th position}}}{1}, 0, \dots, 0]^T$, then $0 = z^T e_i = z_i, \forall i$.
 $\Rightarrow z = 0$

Proof Method 2 (one line): Pick a special x .

$$\text{Pick } x = z, \text{ then } 0 = z^T x = z^T z = \|z\|^2 \Rightarrow z = 0$$

Next: Make It Solvable

P2': find $\mathbf{y} \in \mathbb{R}^{n \times 1}$ such that
 $\mathbf{b} - A\mathbf{y} \perp \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^{n \times 1}\}$

Lemma:

$$u^T x = 0, \forall x \in \mathbb{R}^n \\ \text{iff } u = 0.$$

$$\iff (b - Ay)^T Ax = 0, \quad \forall x \in \mathbb{R}^n$$

$$\iff (b - Ay)^T A = 0$$

$$\iff b^T A - (Ay)^T A = 0 \quad [\text{write as two terms}]$$

$$\iff b^T A = (Ay)^T A \quad [\text{move to another side \& remove bracket}]$$

$$\iff A^T b = A^T (Ay) \quad [\text{take transpose, to make it a more familiar form}]$$

$$\iff A^T b = A^T Ay$$

Check Lec 5,
for rules on transpose.
 $(X Y)^T = Y^T X^T$.

Problem P2' is equivalent to:

P3: find $\mathbf{y} \in \mathbb{R}^{n \times 1}$ such that $A^T A \mathbf{y} = A^T b$

Lemma 2

In the past few pages, we proved the following result.

Lemma 16.2

Suppose $A \in \mathbb{R}^{m \times n}$. Then for any vectors $\mathbf{b} \in \mathbb{R}^{m \times 1}$, $\mathbf{y} \in \mathbb{R}^{n \times 1}$

$$(1) \mathbf{b} - A\mathbf{y} \perp C(A)$$

$$\iff$$

$$(2) A^T A\mathbf{y} = A^T \mathbf{b}$$

Recall: $C(A)$ is the column space of matrix A .

$$C(A) = \{A\mathbf{x}: \mathbf{x} \in \mathbb{R}^{n \times 1}\}$$

Summary of the Journey

Problem P1 find $\mathbf{y}$ s.t. $\|A\mathbf{y} - \mathbf{b}\| \leq \|A\mathbf{x} - \mathbf{b}\|$ for any $\mathbf{x}$ (*)

$\Leftrightarrow$ P1' Find $\mathbf{p} \in \mathcal{C}(A)$ s.t. $\|\mathbf{p} - \mathbf{b}\| \leq \|\mathbf{u} - \mathbf{b}\|$ for any $\mathbf{u} \in \mathcal{C}(A)$

$\Leftrightarrow$ **P2**: find $p \in \mathcal{C}(A)$ such that $\mathbf{b} - \mathbf{p} \perp \mathcal{C}(A)$ (**)

$\Leftrightarrow$ **P2'**: find $\mathbf{y} \in \mathbb{R}^{n \times 1}$ such that

$$\mathbf{b} - A\mathbf{y} \perp \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^{n \times 1}\}$$

$\Leftrightarrow$ **P3**: find $\mathbf{y} \in \mathbb{R}^{n \times 1}$ such that $A^T A\mathbf{y} = A^T \mathbf{b}$

Solving Least Squares: Characterization

The least squares solution can be found by solving a linear system!

Theorem 16.1 (LS solution and normal equation)

Given $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^{m \times 1}$. The following statements are equivalent:

1. $\mathbf{y}$ is a solution of the problem $\min_{\mathbf{x} \in \mathbb{R}^{n \times 1}} \|A\mathbf{x} - \mathbf{b}\|$.

2. $A^\top A \mathbf{y} = A^\top \mathbf{b}$ [This is an $n \times n$ linear system]

Proof: Directly combine two lemmas Lemma 16.1 & Lemma 16.2.

The linear system $A^\top A \mathbf{y} = A^\top \mathbf{b}$ is called the **normal equation**.

Part III.3 Analyzing Normal Equation

3 mins

Solving Least Squares: Normal Equation

Original equation $Ax = b$ may not have a solution.

Question: Does $A^T A \mathbf{y} = A^T \mathbf{b}$ have a solution?

Intuitively, **projection onto a space exists**;
so a solution of normal equation exists.

Proposition (Existence)

The linear system $A^T A \mathbf{y} = A^T \mathbf{b}$ has at least one solution.

Proof skipped for now.

Hint: Use the following fact:
“ $Bx=z$ ” is solvable iff $z \in \mathcal{C}(B)$.
But there is much more than this fact.

Solving Least Squares: Expression for Full-Rank A

Corollary (Characterization)

Suppose A has **linearly independent columns** (i.e., has full column rank).

Then solution of the least square problem $\min_{\mathbf{x} \in \mathbb{R}^{n \times 1}} \|A\mathbf{x} - \mathbf{b}\|$ is

$$\mathbf{y} = (A^T A)^{-1} A^T \mathbf{b}$$

Remark: A has linearly independent columns $\iff \text{rank}(A) = n$

Proof: **Step 1:** Prove “invertible”.

Lemma: If A has full column rank, then $A^T A$ is invertible.

Proof skipped for now.

Step 2: Solution of $A^T A \mathbf{y} = A^T \mathbf{b}$ is $(A^T A)^{-1} A^T \mathbf{b}$.



Summary Today (write Your Own)

One sentence summary:

Detailed summary:

Summary Today (of Instructor)

One sentence summary:

We have studied least squares and analyzed its solution

Detailed summary:

1. What can we do when a linear system has no solutions?

Solve least squares!

2. Applications: linear regression

e.g. predict salary

3. Characterization of its solution and uniqueness condition

Solution satisfies orthogonality condition.

Solution satisfies the **Normal equation**.

Full-column-rank case: Unique expression $\mathbf{y} = (A^T A)^{-1} A^T \mathbf{b}$

Summary Today (of Instructor)

One sentence summary:

We have studied least squares and analyzed its solution

Detailed summary:

1. What can we do when a linear system has no solutions?

Solve least squares!

2. Applications: linear regression

e.g. predict salary

3. Characterization of its solution and uniqueness condition

Solution satisfies orthogonality condition.

Solution satisfies the **Normal equation**.

Full-column-rank case: Unique expression $\mathbf{y} = (A^T A)^{-1} A^T \mathbf{b}$

Summary Today (of Instructor)

One sentence summary:

We have studied least squares and analyzed its solution

Detailed summary:

1. What can we do when a linear system has no solutions?

Solve least squares!

2. Applications: linear regression

e.g. predict salary

3. Characterization of its solution and uniqueness condition

Solution satisfies orthogonality condition.

Solution satisfies the **Normal equation**.

Full-column-rank case: Unique expression $\mathbf{y} = (A^T A)^{-1} A^T \mathbf{b}$

Summary Today (of Instructor)

One sentence summary:

We have studied least squares and analyzed its solution

Detailed summary:

1. What can we do when a linear system has no solutions?

Solve least squares!

2. Applications: linear regression

e.g. predict salary

3. Characterization of its solution and uniqueness condition

Solution satisfies orthogonality condition.

Solution satisfies the **Normal equation**.

Full-column-rank case: Unique expression $\mathbf{y} = (A^T A)^{-1} A^T \mathbf{b}$

Summary Today (of Instructor)

One sentence summary:

We have studied least squares and analyzed its solution

Detailed summary:

1. What can we do when a linear system has no solutions?

Solve least squares!

2. Applications: linear regression

e.g. predict salary

3. Characterization of its solution and uniqueness condition

Solution satisfies orthogonality condition.

Solution satisfies the **Normal equation**.

Full-column-rank case: Unique expression $\mathbf{y} = (A^T A)^{-1} A^T \mathbf{b}$